Examiner only

6. Jamie is investigating series and parallel electrical circuits.

The following equations apply to electrical circuits:

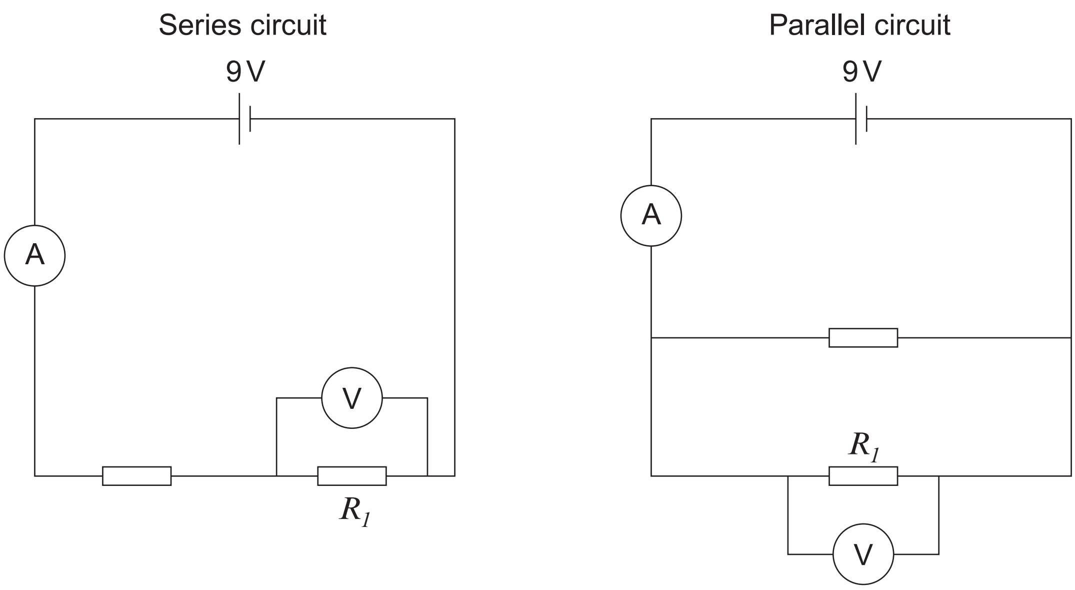
$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

$$\text{power} = \text{voltage} \times \text{current}$$

$$\text{power} = \text{current}^2 \times \text{resistance}$$

$$R_T = R_1 + R_2 \ \ldots\ldots\ldots\ldots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} \ \ldots\ldots\ldots\ldots$$



(a) Jamie built the circuits above with identical resistors. In the series circuit the total resistance in 4.5 ohms and the ammeter reading is 2 Amps.

(i) I. Compare the total resistance in the parallel circuit with the series circuit. [2]

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II. Compare the ammeter reading in the parallel circuit with the series circuit. [1]

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(ii) Explain how the voltmeter readings in the two circuits compare. [2]

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(iii) Explain how the power dissipated in resistor R_1 in each circuit compares. [2]

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(b) Jamie used identical 2.25 Ω resistors in both circuits.

(i) Calculate the power dissipated by R_1 in the parallel circuit. [4]

power dissipated = ... W

(ii) Use your answer above to calculate the power dissipated in resistor R_1 in the series circuit. [2]

power dissipated = ... W

13